AI Sales Automation – System Design & Implementation Approach

Purpose
This document describes how the proposed AI Sales Automation system will be implemented at a high level. It explains what each agent does, what inputs it consumes, how decisions are made, and why certain agents have been merged. The intent is to provide clarity on system behavior before implementation begins.

Agent A: Immediate Call Intelligence Agent

Scope
Agent A runs immediately after every inbound or outbound sales call ends. It consolidates the responsibilities of Lead Qualification & Routing, CSR Booking Optimization, and Rep Performance Coaching.

Inputs
• Call transcript
• Call metadata (duration, speaker turns, call outcome)
• CRM context (lead source, prior interactions)

Lead Priority Scoring
Each lead is assigned a priority score immediately after the call. The score is calculated using four pillars:

1. Intent Signals – What the customer explicitly says (e.g., asking about price, booking, timelines).
2. Urgency Signals – When the customer wants to act (e.g., today, this week, later).
3. Fit Signals – Whether the lead matches service criteria (location, budget cues, decision-maker).
4. Behavioral Signals – Engagement indicators such as call duration, follow-up questions, responsiveness.

Example:
A customer asking for pricing and availability within the same week, actively asking questions, will receive a high score. A customer saying they are only exploring options with no urgency will receive a lower score.

Based on this score, leads are sorted and routed:
• High score → immediate CSR action
• Medium score → follow-up or nurture
• Low score → automated nurture sequence

Booking Failure Detection
Agent A uses an LLM with a structured evaluation prompt to analyze the transcript and

determine whether a booking attempt was made and why it failed. Failure reasons include weak urgency creation, unhandled objections, or lack of a clear next step.

For unbooked leads with sufficient intent, Agent A automatically sends contextual follow-up messages to the customer and creates rescue tasks for CSRs.

Rep Performance & Coaching
Using the same transcript, Agent A evaluates the CSR against predefined SOPs. It identifies behavior-specific issues such as missed objections or poor closing attempts and generates targeted coaching tasks. Failure detection is reused from the booking analysis, ensuring no duplicate logic.

Agent B: Lifecycle Scheduler Agent

Scope
Agent B handles all time-based actions in the system. It merges Stalled Deal Rescue and Appointment Show Rate & Rep Preparation.

Inputs
• CRM deal status
• Proposal timestamps
• Appointment schedules
• Intelligence outputs from Agent A

Stalled Deal Detection
Agent B periodically checks for deals where a proposal has been shared but no response has been received within a defined time window. Once detected, it uses historical call context and objections identified earlier to classify the most likely stall reason.

Based on the detected objection (price, timing, competitor, decision-maker), Agent B sends contextual automated follow-ups to the customer and smart nudges to the assigned rep.

Appointment Reminders & Rep Preparation
For confirmed appointments, Agent B sends automated reminders to customers at predefined intervals to reduce no-shows. Prior to the appointment, it generates a concise preparation brief for the rep, highlighting key customer concerns, likely objections, and recommended talking points.

Agent C: Manager Shadow (Orchestration)

Scope
Agent C acts as the orchestration and oversight layer of the system. It remains separate from other agents to ensure stability and clear governance.

Inputs
• Outputs and events from Agent A and Agent B
• Team workload and capacity data
• System-wide performance metrics

Responsibilities
• Monitor lead movement and agent decisions
• Balance workload across CSRs
• Escalate high-risk or high-value situations
• Generate weekly performance summaries for management

Agent C primarily relies on deterministic rules and thresholds, using LLMs only for summarization and insight generation in reports.

Why This Architecture

Agents are merged based on execution timing and shared data dependencies. Immediate post-call logic is consolidated into Agent A, time-based lifecycle actions into Agent B, and system governance into Agent C. This reduces complexity while preserving all original capabilities.

Next Steps

Following approval, the implementation phase will begin with workflow design, prompt finalization, and system integration planning.